# Experiment - 3

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## 1.COMMAND LINE INTERFACE:
## PYTHON CODE:

```
import sys
from datetime import date

if len(sys.argv) < 3:
    print("Usage: python age_cli.py <birth_year> <birth_month>")
    sys.exit()

birth_year = int(sys.argv[1])
birth_month = int(sys.argv[2])

today = date.today()
age = today.year - birth_year

if today.month < birth_month:
    age -= 1

print("Your age is:", age)
```

OUTPUT:

```
PS C:\Users\Pavithra\OneDrive\Desktop\File Remaining System> python cli.py 2003 8
>> & C:/Users/Pavithra/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/Pavithra/OneDrive/Desktop/File Remaining System/cli.py"
Your age is: 22
Usage: python age_cli.py <birth_year> <birth_month>
PS C:\Users\Pavithra\OneDrive\Desktop\File Remaining System>
```

## 2.GRAPHICAL USER INTERFACE:

## PYTHON CODE:

```
import tkinter as tk
from datetime import date

def calculate_age():
    birth_year = int(entry_year.get())
    birth_month = int(entry_month.get())

    today = date.today()
    age = today.year - birth_year

    if today.month < birth_month:
        age -= 1

    label_result.config(text=f"Your age is: {age}")

root = tk.Tk()
root.title("Age Calculator")

tk.Label(root, text="Birth Year").pack()
entry_year = tk.Entry(root)
entry_year.pack()

tk.Label(root, text="Birth Month").pack()
entry_month = tk.Entry(root)
entry_month.pack()

tk.Button(root, text="Calculate Age", command=calculate_age).pack()

label_result = tk.Label(root, text="")
label_result.pack()
```

```
root.mainloop()
```

OUTPUT:

## 3.VISUAL USER INTERFACE:

PYTHON CODE:

```
import speech_recognition as sr

def listen(prompt):
    recognizer = sr.Recognizer()
    with sr.Microphone() as source:
        print(prompt)
        audio = recognizer.listen(source)
        try:
            text = recognizer.recognize_google(audio)
            print(f"You said: {text}")
            return text
        except sr.UnknownValueError:
            print("Sorry, I did not understand. Please try again.")
            return listen(prompt)
        except sr.RequestError:
```

```
        print("Could not request results; check your internet connection.")
        exit()

birth_year = int(listen("Say your birth year:"))
current_year = int(listen("Say the current year:"))
age = current_year - birth_year
print(f"Your age is: {age}")
feedback = listen("Are you satisfied with the result? Say 'yes' or
'no':").strip().lower()

if feedback == "yes":
   print("Thank you for your feedback ")
elif feedback == "no":
   print("We will try to improve ")
else:
   print("Invalid input, please say 'yes' or 'no'")
```

OUTPUT:

```
PS C:\Users\Pavithra\OneDrive\Desktop\Age Calculator file> & C:/Users/Pavithra/AppData/Local/Programs
/Python/Python313/python.exe "c:/Users/Pavithra/OneDrive/Desktop/Age Calculator file/vui.py"
Say your birth year:
You said: 2006
Say the current year:
You said: 2026
Your age is: 20
Are you satisfied with the result? Say 'yes' or 'no':
Sorry, I did not understand. Please try again.
Are you satisfied with the result? Say 'yes' or 'no':
Sorry, I did not understand. Please try again.
Are you satisfied with the result? Say 'yes' or 'no':
Sorry, I did not understand. Please try again.
```

```
Are you satisfied with the result? Say 'yes' or 'no':
You said: no
We will try to improve
```

## 4.USER SATISFACTION INTERFACE:

## PYTHON CODE:

```
birth_year = int(input("Enter your birth year: "))
current_year = int(input("Enter the current year: "))

age = current_year - birth_year
print(f"Your age is: {age}")

feedback = input("Are you satisfied with the result? (yes/no): ").strip().lower()

if feedback == "yes":
   print("Thank you for your feedback ")
elif feedback == "no":
   print("We will try to improve ")
else:
   print("Invalid input, please type 'yes' or 'no'")
```

OUTPUT:

```
PS C:\Users\Pavithra> & C:/Users/Pavithra/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/Pavithra/OneDrive/Desktop/Age Calculator file/user_satisfaction.py"
Enter your birth year: 2006
Enter the current year: 2026
Your age is: 20
Are you satisfied with the result? (yes/no): yes
Thank you for your feedback
PS C:\Users\Pavithra> & C:/Users/Pavithra/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/Pavithra/OneDrive/Desktop/Age Calculator file/user_satisfaction.py"
Enter your birth year: 2004
Enter the current year: 2026
Your age is: 22
Are you satisfied with the result? (yes/no): no
We will try to improve
```